

Adapting an Ecological Belonging Intervention to an Introductory Computer Science Course

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Background and Context. Prior research has shown that students' sense of belonging and self-efficacy are linked to their long-term performance and persistence in the computer science (CS) major. As a result, efforts specifically targeting these metrics may serve to broaden participation in CS. Brief psychological interventions within introductory classrooms have shown wide success in boosting student performance and sense of belonging in many other STEM fields, but the few implementations in CS courses have shown only mild success. More recently, STEM Education researchers developed an Ecological Belonging Intervention (EBI) that targets social contexts rather than individual psychology. It aims to create ecological change by introducing the classroom norm that adversity is surmountable and not a result of fixed or internal causes. The intervention was run in introductory biology, physics, and engineering classrooms, where it closed gender-based and race-based gaps in both performance and sense of belonging, and improved retention for students from historically underrepresented racial groups.

Objectives. We sought to determine whether the EBI has similar effects in an introductory CS classroom. Additionally, we investigated the EBI's impact on students' self-efficacy.

Method. We conducted a between-subjects study in an introductory CS course intended for students without prior programming experience, the majority of whom were women or non-binary. We evaluated the EBI's influence on students' sense of belonging, self-efficacy, performance, engagement, and persistence. Additionally, we examined whether the EBI is more effective for students from specific demographic groups. We measured students' sense of belonging in CS using a recently developed five-factor instrument that measures a student's *Acceptance, Affect, Membership, Desire to Participate, and Trust*. We measured students' self-efficacy using the corresponding subscale in the Motivated Strategies for Learning Questionnaire (MSLQ) instrument.

Findings. We found that students who received the EBI experienced significantly larger increases in self-efficacy than students who did not. Moreover, first-year students who received the EBI had significantly larger increases in the *Affect* factor of their sense of belonging. There were no other significant effects between the EBI and the outcomes of either the entire population or specific demographic groups.

Implications. The EBI's ability to improve students' self-efficacy and the *Affect* factor of their sense of belonging suggests that it has the potential to increase CS students' persistence and performance *over time*. We encourage researchers to further explore the EBI's benefits in other CS contexts.

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1 Introduction

Prior research in **Computer Science (CS)** education has shown that self-efficacy and sense of belonging are related to students' long-term performance [25] and persistence [9, 44, 50] in CS. However, students from **Historically Underrepresented Groups (HUGs)** in CS, including women, non-binary, Black/African-American, Hispanic, Latinx/e/a/o, and Native American students [4, 19, 29, 34], tend to have lower self-efficacy [13, 28, 35, 46, 48] and sense of belonging [13, 16, 25, 41, 48].

To tackle this phenomenon, researchers in other STEM fields have recently employed an **Ecological Belonging Intervention (EBI)** [10, 11, 21] (see Section 2.4 for an in-depth description) in introductory STEM courses. This intervention introduces as a classroom norm that the challenges students are facing in transitioning to college and in their courses are normal and can be overcome. While the EBI has been adapted to engineering programming contexts, where it improved retention and sense of belonging for Black, Latinx/e/a/o, and Indigenous students [6, 20], it has not yet been studied in CS classrooms. Additionally, no prior works have explicitly investigated the EBI's influence on computing students' self-efficacy.

In this study, we adapt the EBI to an introductory CS course to determine the extent to which the EBI is beneficial in a CS context. This course was intended for students without prior programming experience, many of whom were women or non-binary. We evaluate (RQ1) how the EBI influences students' sense of belonging, self-efficacy, exam performance, engagement, and persistence. Furthermore, we examine (RQ2) whether the intervention is more effective for certain demographic groups. Specifically, we consider students' gender, racial/ethnic, first-generation, and year/standing identities, as well as prior programming experience.

We found that students who received the EBI experienced significantly larger increases in self-efficacy (Section 6.1). Additionally, among first-year students, those who received the intervention had significantly larger increases in the *Affect* factor of their sense of belonging (Section 6.2). We found no significant effects between the intervention and students' performance on exams, course engagement, persistence, and other aspects of sense of belonging. We discuss the implications of our findings in Section 7.

2 Related Work

2.1 Psychological Sources of Inequity in CS

The presence of psychological factors that drive inequity in CS is well-documented. For conciseness, we only reference a sampling of works relevant to our research questions. For a broader overview, we direct the reader to the following literature reviews by Malmi et al. [31] and Batra and Atiq [8] studying psychological factors in CS Education, as well as the literature reviews by Shehzad et al. [43] and Agbo [2] about efforts to broaden participation in CS among elementary school and adult learners, respectively.

Many studies have shown that students from HUGs in CS face emotional and affective barriers to persistence. Hunt et al. [23] show that women in CS have a lower self-assessment of their abilities than men with equivalent grades, and that this lower self-assessment is tied to lower long-term persistence in CS. Sax et al. [41] and Krause-Levy et al. [25] find that women, students from under-represented minority races and ethnicities, first-generation students, and transfer students all have a lower sense of belonging in CS courses. They also find that this lower sense of belonging is tied to lower course performance, with the correlation being strongest in introductory CS courses. Ojha et al.'s multi-institutional study [35] reports that prior programming experience positively predicts CS students' self-efficacy, while identifying as a woman or non-binary and identifying as Black, Native, Hispanic, or Asian both negatively predict self-efficacy.

Several well-known psychological threats have been demonstrated to affect CS students as well. Rosenstein et al. [40] find that impostor phenomenon is higher in CS than in other fields and that women in CS exhibit higher levels of impostor phenomenon than men. Kumar [27] finds that average-performing and below-average-performing female students performed worse when reporting their sex before an assessment on loops, demonstrating the existence of stereotype threat in introductory CS courses.

2.2 Social-Psychological Interventions

Within the field of education, social-psychological interventions—interventions that directly target student beliefs and thoughts—have emerged as a potent way to address psychological threats to students [17]. This is in contrast to interventions that make changes to course materials or policy with the goal of indirectly improving student affect. Social-psychological interventions tend to be brief but demonstrate lasting benefits, especially for students from HUGs. For example, Walton and Cohen [49] find that a 60-minute intervention, in which students were shown evidence that social adversity is both common and temporary, improved academic and health outcomes of African-American first-year students. To explain the underlying psychological mechanism, the authors note that in follow-up surveys, African-American students in the treatment and control conditions report facing similar levels of adversity, but that adversity was only correlated with a sense of belonging among students in the control group. This indicates that the intervention buffered students from attributing the hardships they faced to their ethnicity. Despite the significant effect, treatment students also report no awareness of the intervention's impact. A follow-up study finds that academic and health benefits persist a decade later [14]. Similarly, a 90-minute intervention that engaged students in a discussion about social belonging reduced the risk of major depression and self-reported experiences of discrimination among **Black, Indigenous, and People of Color (BIPOC)** and first-generation college students, as studied over a two-year period [42].

Benefits from social-psychological interventions have been shown to extend to all students, not just students in minority groups: Powers et al. [38] find emergent environmental benefits from a social-psychological intervention in that the proportion of treated students in a class predicted individual students' performance, no matter the student's race or whether that student was treated.

In a seminal review of social-psychological interventions, Yeager and Walton [53] argue that despite the brief nature of these interventions and their comparatively large effects, they should not be considered "magical" or "silver bullets." Instead, they argue that these interventions have lasting effects

"because they target students' subjective experiences in school, because they use persuasive yet stealthy methods for conveying psychological ideas, and because they tap into recursive processes present in educational environments."

We provide a more detailed description of each of these factors in Sections 2.2.1–2.2.3. The review by Yeager and Walton describes how social-psychological interventions do not necessarily immediately revolutionize student experiences, but instead spur small changes in the processing of psychological threats that become self-reinforcing over a student’s career. As such, they caution that scaling these interventions to different students requires careful consideration of how the underlying mechanisms and theories interact with specific contexts.

2.2.1 Targeting Students’ Subjective Experiences in School. Yeager and Walton emphasize that these experiences must be centered on students’ subjective experiences in school, rather than an instructor or researcher’s view of the environment. While the 60-minute intervention about social adversity may appear to be a small fraction of a day for the instructor, it can be enough to have an outsized impact on students who are currently facing said adversity. To that end, Yeager and Walton encourage researchers to be deliberate about the length of the interventions: while one might assume that an increased length leads to an increased impact, students may perceive a more heavy-handed intervention as a sign that they need more help or as taking away more from the core course content.

2.2.2 Persuasive yet Stealthy Methods for Conveying Psychological Ideas. For a similar reason, Yeager and Walton emphasize the importance of using “stealthy” methods, in the sense that students should not realize they are part of an intervention. For example, in Walton and Cohen’s intervention [48, 49], students were asked to act as mentors for younger students, and they were presented with statistics normalizing social adversity under the guise that the statistics would help them develop materials for younger students. In this sense, the students needed to critically engage with the provided statistics and develop reassuring advice for “younger students,” unaware that they themselves were the target audience of the advice.

2.2.3 Tapping into Recursive Processes Present in Educational Environments. As we described earlier, the brief and stealthy nature of these interventions may make their large observed impacts seem implausible. To explain this, Yeager and Walton credit recursive processes—ways in which the original intervention may be self-reinforcing through the student’s academic career. For example, if a student internalizes an intervention’s message and maintains a higher self-efficacy in the face of a bad assessment score, then they are more likely to be willing to put effort into preparing for the next assessment. This effort is likely to lead to better performance, which leads to even higher self-efficacy and an increased belief in the intervention message, and the cycle repeats. This self-reinforcing loop describes how the task of the initial intervention is simply to help students gain the momentum to create positive outcomes for themselves. This gives students the psychological security that they need to keep making efforts to succeed. This explains why Brady et al. [14] found that students who received the intervention from Walton and Cohen’s work [49] benefited significantly in the long-run even though very few of them could recall the contents of the intervention.

2.3 Interventions to Close Gaps in CS

Despite the wide success of social-psychological interventions in STEM, few studies have implemented these interventions in CS courses. We find two instances in which different studies implement an intervention targeted at helping students develop a growth mindset. The dearth of further work may be in part because these interventions were not conclusively beneficial. Krause-Levy et al. [26] simultaneously implement three interventions in an introductory CS course, one of which is a growth mindset intervention, and find only “mild positive effects for some students.” Burnette et al. [15] conduct a larger, multi-institutional study on a growth mindset intervention and find that the intervention increased students’ perceived intrinsic value of the course, but not

their performance. We also find a storytelling-based social-psychological intervention conducted by Reckinger et al. [39] in an upper-level systems CS course. They find no statistically significant changes in students' sense of belonging or retention. They also find that students in the control group have a strong correlation between their sense of belonging and enjoyment in the course, but students in the experimental group do not. This may reflect their ability to separate their enjoyment of a course from their sense of belonging. These works complicate the discussion about the effectiveness of psychological interventions in CS, making it more important to continue studying them in our field.

We note that *social-psychological interventions* are a narrower class of interventions than *interventions that affect social and psychological factors*. A social-psychological intervention is *directly targeted at changing student beliefs*, differentiating it from interventions changing other course aspects like course policy (e.g., [34]), curriculum (e.g., [24]), or feedback (e.g., [3]), with the end-goal of improving the factors of interest. Though we only find three studies describing social-psychological interventions in CS courses, many studies have conducted interventions that aim to indirectly improve these gaps, as described in the literature review in Section 2.1.

2.4 The EBI

In this study, we adapt the EBI, a social-psychological intervention developed by Binning et al. [11], to an introductory CS course. Their work notes that psychological threats can be interpersonal, and choose to target classroom environments instead of only individual beliefs. Specifically, they aim to alter the classroom's social context by introducing the classroom norm that adversity is surmountable and not a result of fixed or internal causes, such as an individual's race or gender. The authors argue that this norm would encourage *all* students to attribute differences in performance to "universal and temporary causes" rather than competence stereotypes, thereby targeting interpersonal interactions as well. They conducted the EBI in college biology and physics courses, where they find that students who participated in the intervention earned higher course grades, had greater attendance in discussion sections, and were more likely to persist in college for an additional year than students who had not. The intervention also eliminated performance gaps between White and non-White students in the biology course and reduced performance gaps between men and women in the physics course. In a five-year follow-up work [10], they find evidence that the EBI successfully changed the classroom ecology. Specifically, the EBI reduced the gender gap in sense of belonging and increased cross-gender collaboration. However, they also find that within classrooms with low gender diversity (<25% women), women who experienced the EBI had lower growth mindsets than women who had not. The authors hypothesize that this occurred because the social context of the course contradicted the intervention message. This finding emphasizes the importance of studying how differing contexts can affect intervention outcomes.

Continuing this work, Hammerlund et al. [21] sought to determine when the EBI was most effective at eliminating performance gaps between marginalized and non-marginalized students. They conducted the intervention in two non-major courses. The first was an introductory chemistry course, where students who identified as American Indian, Black, Hawaiian, and Hispanic tended to underperform. The second was an introductory biology course, where first-generation students tended to underperform. Their results suggest that the EBI is most effective in contexts that may be threatening to students. The intervention improved the performance of (1) American Indian, Black, Hawaiian, and Hispanic students in the chemistry course, and (2) students in the biology course whose **Teaching Assistants (TAs)** had a fixed mindset. However, the intervention did not improve the performance of first-generation students in the biology course. The authors hypothesized that this was because first-generation status is not visible, and therefore may not

Table 1. Demographics of *Consenting Students*

# Total students	Consenting (%)	Gender		Race				Latinx/e/a/o		FirstGen		Year			
		H-Min	H-Maj	White	Asian	Black	2+	Y	N	Y	N	1	2	3	4+
199	159 (79.9%)	83	45	61	39	13	5	25	101	38	121	67	41	10	9

H-Min includes women, non-binary, and transgender students. Black includes African-American students; 2+ means multiracial. Not all students answered all questions. Shaded cells for *Race* and *Year* are merged in our analysis.

have been discussed during the intervention unless a student volunteered that information to their peers. Our study considers first-generation status and other invisible identities, including prior programming experience, so our study may help determine whether the visibility of a student’s identity affects the impact of the intervention.

To the extent of our knowledge, the EBI has not been studied in introductory CS courses prior to our work. The closest setting to ours is a series of papers by Bahnson, Godwin, and others [5, 6, 20] that study the effect of the EBI on an introductory engineering programming (MATLAB) course in Spring 2022 and Spring 2023. In Spring 2022, they group **Black, Latinx, and Indigenous (BLI)** students together and find a three-way interaction effect between treatment, BLI membership, and time (before vs. after the course) on sense of belonging [20]. In Spring 2023, they find that for BLI students, (1) the treatment group had significantly higher retention in engineering, and (2) belonging increased in both control and treatment groups but only predicted retention in the treatment group [6]. In a follow-up study, they find that belonging increased for women and that this was associated with increased retention [5].

2.5 Our Contributions

Our work adapts the EBI to an introductory CS context to determine whether the EBI can have a positive influence on computing students. Following Binning et al. [10, 11], we examine the EBI’s effects on students’ performance, sense of belonging, course engagement, and major retention. However, we also expand on their work by (1) looking into whether the EBI influences students’ self-efficacy, and (2) using a more fine-grained belonging instrument with evidence of validity.

3 Context

Our data is from the Fall 2024 offering of an introductory programming course (IntroPython) at a medium-sized, research-oriented private university with 15-week semesters. Student demographics are summarized in Table 1.

The course is not required for the CS major or minor and is meant for students without prior programming experience. Consequently, it is only open to students who have not yet taken any programming courses at the university. Students who have other forms of prior programming experience are strongly encouraged to instead take a more advanced Data Structures and Algorithms course. This is likely why most of the students in our data had no (35%) or little (47%) prior programming experience (see Table 2) and identified as being from a **Historical Gender Minority (H-Min)** in computing (see Table 1).

Most students take this course to determine whether they would be interested in majoring in CS, or because they intend on majoring in CS but did not take a programming course in high school. However, since IntroPython is a required course for both Statistics and Neuroscience majors, some students also take this course to fulfill these requirements. As students at our institution cannot declare their major until their second year, many students in our data were undeclared at the time of taking this course. We thus do not report students’ majors in Table 1.

Table 2. Distribution of Prior Programming Experience (*PriorExp*) among Consenting Students

Group	Prior programming experience level	# Students	(%)
No <i>PriorExp</i>	"I have/had never written a line of code in my life."	56	35.2%
Some <i>PriorExp</i>	"I have/had played around with coding a bit, but never written anything beyond a few lines. I don't know how to/feel confident writing loops or if statements."	74	46.5%
	"I have/had written multiline programs, using ifs and loops to build a function or method."	26	16.4%
	"I have/had written many large programs combining multiple functions/methods."	3	1.9%

All 159 consenting students responded to a multiple-choice question about their level of prior programming experience. The second column contains the options students could choose from.

3.1 Course Structure and Deliverables

The course met three times a week. The first two meetings per week were lectures, during which the instructor introduced new concepts to the students. The instructor incorporated in-class activities into lectures and asked students to respond to online forms as part of the activities. They then used students' responses to the online forms as a proxy for lecture attendance. Students earned 5% of their grade by responding to 75% of all forms across the semester. The third meeting was a lab section, during which students completed a series of exercises that focused on concepts taught in lecture. Lab sections were capped at 27 students and run by undergraduate TAs. Attendance was required, although students were allowed to miss up to three labs without penalty.

The course had two midterm exams and one final exam. Each exam was worth 18% of students' final grades. However, some students opted to not take the final exam because they were taking the course on a pass/fail basis and knew that they could pass with a 0% on the final exam.

All course assignments were autograded and formative in nature, and students were allowed to resubmit them an unlimited number of times before the deadline. Consequently, there was minimal variation in assignment performance, with most students receiving full credit. We therefore do not examine students' assignment grades in this study.

3.2 Course Environment

Our institution does not admit undergraduate students by major and does not have competitive enrollment for the CS major. Specifically, all students who apply to be a CS major are accepted, regardless of their GPA, prior coursework, or extracurricular activities. This mechanism may be tied to a higher sense of belonging and self-efficacy, especially among students with no prior experience [33]. Moreover, women are *not* the minority in this course (see Table 1). These factors differentiate our course climate from that of prior works on the EBI and introductory CS courses at many other institutions.

Moreover, the instructor designed certain aspects of the course with the intention of making the course environment more welcoming. For example, they highlighted a computer scientist with a historically underrepresented identity at the beginning of each lecture. This was meant to, in their words, "dismantle the stereotype that the white man is the only person doing CS."

4 Study Design

The intervention was modeled after that of Binning et al. [11] and approved by our IRB. It was conducted in five stages, totaling 30 minutes, and aimed to establish the classroom norm that hardships in transitioning to college are normal and surmountable. The intervention consisted of individual reflection activities and group discussions, as outlined in Table 3. A more detailed description of the intervention can be found in Appendix A.

Table 3. Outline of the EBI and the Control Activity

Phase	Intervention	Control
Activity Introduction (3 minutes)	Students are told: – The transition to college is difficult – Everyone experiences self-doubt at times – Things improve over time	Students are told: – College workload is larger – Things get easier with good study habits
Individual Writing and Reflection (7–10 minutes)	Students reflect on: – Challenges they faced since entering college – How these challenges improved over time	Students reflect on: – Study habits that have and haven't worked for them in the past
Reading Quotes From "Prior" Students ^a (5–7 minutes)	Quotes describe: – Adversity is normal and not unique to certain demographic groups – Challenges are surmountable and get easier – Learning from one another	Quotes describe: – Improving study habits – Utilizing available course resources – Individually overcoming challenges
Small Group Discussion of Reflection and Quotes (7–10 minutes)	Discussion questions presuppose hardship: e.g., <i>"Why do you think that sometimes students don't realize that other people are also struggling with the course?"</i>	Discussion questions center studying: e.g., <i>"What are some common study habits that have helped other students?"</i>
Whole Class Discussion (3–5 minutes)	– Students share from group discussion – TA reiterates message that struggles are normal and temporary	– Students share from group discussion – TA includes references to personal study habits as is appropriate

^aQuotes were written by researchers in Binning et al. [11] and adapted for our course context. See Appendix A.

4.1 Logistics

The intervention took place during the first lab section of the Fall 2024 semester. It was integrated into existing lab materials and framed as a "reflection activity" as opposed to a social-psychological intervention so as to not discourage buy-in [53].

There were 10 lab sections, half of which were randomly assigned to receive the intervention. Following the procedure by Binning et al. [10, 11], we did not use a standard control where students in the control group do nothing. Rather, students in the control sections participated in an alternative activity that has been shown not to have effects when used as a control [12]. This second activity had the same structure as the intervention but asked students to reflect on their study habits and how they intended to learn the course material. It therefore targets individual beliefs as opposed to ecological ones. We note that the use of a standard control group is unrealistic due to the fact that the class time taken up by the EBI cannot be left empty or used to cover additional course material.

Like in Hammerlund et al. [21], the interventions were led by undergraduate TAs, which can increase credibility because they experienced the transition to college at the same institution. To ensure consistency in the delivery of the intervention, all 10 sections were led by the same 4 TAs, who each had a minimum of two semesters of TA experience. Since multiple lab sections occurred at the same time, it was not feasible for the interventions to be run by a smaller number of TAs. The TAs attended a 30-minute training session conducted by the first author. The first author walked the TAs through the contents of the intervention and showed them the scripts they would be using,

which were adapted from Binning et al. [10, 11].¹ In addition, the first author encouraged the TAs to practice delivering the script in advance and emphasized the importance of following the script closely.

5 Methods

5.1 Non-Psychometric Data Collection and Processing

Students filled out a consent form during the first week of classes. The consent form told students that their data would be collected to learn more about instructional effectiveness. We informed students that their responses to the consent form would never be shown to the instructor and would not impact their grade. There was nothing in the consent form to indicate that an intervention was happening; at most, students would be aware that the study has a focus on sense of belonging, self-efficacy, performance, engagement, and persistence. Students were not told that there were two different activities happening in the first lab, and they did not know whether they were part of the intervention or control group. 159 of the 199 students in the course consented to participate in the study, for a consent rate of 79.9%. Among the 159 consenting students, 73 received the intervention; the remaining 86 were in the control group. We describe the different kinds of non-psychometric data collected below.

Students' Demographic Data. This data was collected from a beginning-of-semester survey (*Prior-Exp* and *FirstGen*) available during Week 1 and a mid-semester survey (*Gender*, *Race*, *Latinx/e/a/o*, and *Year*) available during Weeks 9 and 10. We report student demographics in Table 1. In our analysis, we collapsed demographic groups for *Race* and *Year*. We merged the Black and Multiracial (2+) racial groups because both were too small for the analysis of intervention effects. Since the intervention is intended for students who are transitioning to college, we classified second-year, third-year, and fourth-year students as *upperclassmen* (2+) and compared the intervention's effects on first-year students to its effects on upperclassmen.

Students' Midterm and Final Exam Scores. We used this data as a measure of *Performance*. Four students chose to not take the final exam and were excluded from our analyses of final exam scores.

Students' Attendance in Weekly Lab Sections and Participation in In-Class Activities. We used these two datasets as measures of *Engagement*. There were a total of 12 labs during the semester. However, one of them was canceled for more than half of the sections due to inclement weather. As a result, our analysis only included attendance records from the other 11 labs. The instructor collected in-class activity data on the *week* level, so we evaluated students' lecture attendance at the week level. We considered students to have attended a lecture during a specific week if the number of in-class activities they participated in (i.e., the number of forms they submitted) was not 0.

5.2 Psychometric Data Collection and Processing

Belonging. The 26-item *Belonging* instrument (see Table 4) was first adapted by Moudgalya et al. [32] from the math context to the introductory CS context. It was then recently refined by Werner and Chen [51]. This instrument subdivides students' senses of belonging into five factors (constructs): *Acceptance*, *Affect*, *Desire to Participate*, *Membership*, and *Trust*. A 7-point Likert scale ranging from "Strongly Disagree" to "Strongly Agree" was used for all items. Following the analyses in Werner and Chen's work [51], students' responses to each item were coded numerically from 1 (Strongly Disagree) to 7 (Strongly Agree). Some items (see Table 4) were negatively worded and therefore reverse-coded for analysis. A student's score for each construct is the sum of their responses to all the items under that construct in Table 4.

¹Only minor changes were made, to tailor them to our institution.

Table 4. Items in the Survey Instruments

Construct	Item Stem	Item #
<i>Acceptance</i>	I feel respected.	2
	I feel disregarded. ^a	5
	I feel neglected. ^a	14
	I feel valued.	16
	I feel excluded. ^a	18
	I feel accepted.	19
	I feel insignificant. ^a	20
	I feel appreciated.	24
<i>Affect</i>	I feel nervous. ^a	1
	I feel calm.	3
	I feel comfortable.	4
	I feel at ease.	7
	I feel tense. ^a	17
	I feel anxious. ^a	21
<i>Desire to Participate</i>	I enjoy being an active participant.	11
	I wish I could fade into the background and not be noticed. ^a	13
	I try to say as little as possible. ^a	25
	I wish I were invisible. ^a	26
<i>Membership</i>	I feel that I belong to the CS community.	8
	I feel a connection with the CS community.	9
	I consider myself a member of the CS world.	12
	I feel like I am part of the CS community.	22
<i>Trust</i>	Even when I do poorly, I trust my instructors to have faith in my potential.	6
	I trust the testing materials to be unbiased.	10
	I have trust that I do not have to constantly prove myself.	15
	I trust my instructors to be committed to helping me learn.	23
<i>Self-Efficacy</i>	I believe I will receive an excellent grade.	27
	I'm certain I can understand the most difficult material presented in the readings.	28
	I'm confident I can understand the basic concepts taught.	29
	I'm confident I can understand the most complex material presented by the instructor.	30
	I'm confident I can do an excellent job on the assignments and tests.	31
	I expect to do well.	32
	I'm certain I can master the skills being taught.	33
	Considering the difficulty of this course, the teacher, and my skills, I think I will do well.	34

The *Belonging* part (first 5 constructs) is reproduced from Table 5 in Werner and Chen's work [51]. The *Self-Efficacy* construct is from the *Self-Efficacy for Learning and Performance* scale component in the MSLQ [37]. For brevity, we adapted the component by mentioning the "in this class/course" modifier at the beginning of the instrument instead of in each item.

^aReverse-coded items.

Self-Efficacy. We measured students' *Self-Efficacy* using the *Self-Efficacy* subscale in the **Motivated Strategies for Learning Questionnaire (MSLQ)** instrument [37]. We made a minor adaptation to the subscale that was intended to reduce the length of the questions. We describe the change in the caption of Table 4. For both *Belonging* and *Self-Efficacy*, the pre-intervention responses were captured in the beginning-of-semester survey, and the post-intervention responses were captured in an end-of-semester survey released during the last week of classes, prior to the final exam. Although 159 students completed both pre-intervention instruments, only 113 did so post-intervention. As a result, $n = 113$ for all analyses of the two instruments.

To confirm that our usage of these instruments (and our slight modifications of the wordings in the *Self-Efficacy* instrument) did not severely compromise the reliability of the measurements,

Table 5. Cronbach's α Reliability Measures for All Instrument Components Compared to Recent Works in Similar Contexts

Construct	Moudgalya et al. [32]	Werner and Chen [51]	Ours (pre-intervention)	Ours (post-intervention)
<i>Acceptance</i>	.92(+)/.91(−)	.903	.843	.854
<i>Affect</i>	.90	.886	.851	.857
<i>Desire to Participate</i>	.83	.783	.772	.798
<i>Membership</i>	.95	.899	.929	.941
<i>Trust</i>	.79	.712	.603	.721
<i>Self-Efficacy</i>	.937 (Wortmann and Varenhold [52])		.931	.944

Note that Moudgalya et al. [32] separates *Acceptance* into positive (+) and negative (−) parts.

we obtained the Cronbach's α reliability measures of each construct in our data. Table 5 compares our values against recent related works in similar contexts (introductory programming [32, 52] or general postsecondary computing students [51]) that used the same instruments. All values except one are considered acceptable (>0.70) or better. The $\alpha = .603$ for *Trust* pre-intervention is considered “moderate” according to Taber's characterizations [45]. We note that the relative reliability of the constructs is similar to the compared works. Specifically, *Trust* has the lowest α values among all *Belonging* constructs in all listed works.

Persistence. Previous research suggests that students' **Intent to Major (ITM)** is predictive of their short-term future enrollment in CS courses and is a more direct predictor of future persistence than any other attitudinal measures, such as *Belonging* and *Self-Efficacy* [22]. Therefore, we use the ITM data as measures of *Persistence*. The ITM data was collected once in the beginning-of-semester survey and again in the mid-semester survey, where students indicated whether they intended to major in CS, were undecided, or intended to major in a different subject. In our analysis, we coded these responses into 1 (intend to), 0.5 (undecided), and 0 (not intend to), respectively, to capture different strengths of students' ITM in CS.

Measuring Changes in Belonging, Self-Efficacy, and Persistence. All of the constructs above were measured once pre-intervention and once post-intervention. As a result, we were able to measure the *changes* in each construct for each student by taking the differences between their pre-intervention and post-intervention metrics (composite scores for all *Belonging* and *Self-Efficacy* constructs, and the simple coding above for *Persistence*). For example, a positive ITM difference indicates that a student's ITM increased, a 0 indicates that a student's ITM was unchanged, and a negative value indicates that a student's ITM decreased. Note that the magnitudes of the difference values therefore indicate the *strengths* of the changes. This allows us to test, for each construct, whether the *changes* were different between students who received the intervention and students who did not, using appropriate tests for each kind of data (see Section 5.3).

5.3 Statistical Methods

We used non-parametric statistical tests for all of our analyses. This is because our psychometric data is not numerical, and our non-psychometric data is not normally distributed. The only metric that might have been normally distributed is exam scores, but Shapiro-Wilk tests confirmed that the distribution was not normal ($p < 10^{-10}$ for each of the three exams).

Since all of our analyses are between-subjects, we used the **Mann–Whitney U test (MWU)** hereafter) for comparing dependent variables between two subpopulations and the **Kruskal–Wallis H test (KWH)** hereafter) for comparing dependent variables across three or more subpopulations.²

²None of the KWH tests we performed were significant, and we therefore performed no *post-hoc* tests.

Table 6. Interpretation of CLES
Values Reproduced from Vargha and
Delaney [47]

If the CLES is within...	The effect is...
(0.44, 0.56)	Negligible
(0.36, 0.44] or [0.56, 0.64)	Small
(0.29, 0.36] or [0.64, 0.71)	Medium
[0.00, 0.29] or [0.71, 1.0]	Large

We applied a Holm–Bonferroni correction [1] across all tests for each of the nine constructs (see Table 8). For all results, we report the **Common Language Effect Size (CLES)** [47] for the intervention group (CLES-I hereafter), i.e., the proportion of all possible cross-group pairs of students in which the student in the intervention group has a higher metric value than the student in the control group. We interpret all effect sizes using the guidelines set forth by Vargha and Delaney [47], reproduced in Table 6.

5.4 Existing Relationships in the Data

To confirm that there were no unexpected correlations in our data, we ran a Chi-Square test of independence between the experimental conditions and each demographic/identity variable in Tables 1 and 2. None of the tests rejected the independent hypothesis at the 0.05 significance level, which implies that the two treatment groups had similar demographics and prior programming experience. We also conducted a MWU test that compared each of the six constructs in Table 4 between the experimental conditions. No significant results surfaced, which implies that the two treatment groups had similar pre-intervention levels of all six psychometric constructs. We note that *Performance* and *Engagement* were not measured prior to the intervention and therefore could not be compared in this way.

We also attempted to identify existing gaps in the metrics between demographic subgroups. For each construct other than *Engagement*, we conducted a MWU test for each binary demographic variable (*Gender*, *Latinx/e/a/o*, *FirstGen*, *Year*, and *PriorExp*) and a KWH test for *Race*. Because the purpose of this analysis was to identify existing gaps instead of measuring any effects of the intervention, we used the first available absolute measurement for all constructs. The analysis used the pre-intervention scores for all six constructs in Table 4, the scores for the first midterm for *Performance*, and students' ITM in the first week of the semester for *Persistence*.

Table 7 summarizes the significant results. As shown, there were no significant existing relationships regarding *Belonging* and *Self-Efficacy*, and the only post-correction significant results concern *Performance* and *Persistence*. Specifically, *FirstGen* students performed significantly worse than *Non-FirstGen* students on one of the three exams, while first-year students and students with some *PriorExp* have significantly higher ITM than non-first-year students and students with no *PriorExp*, respectively.

Although the effect size for ITM for first-year and non-first-year students was large (CLES-I = 0.80), we note that the *Persistence* effects are likely reflective of the student population of our course: as we mentioned in Section 3, the course serves both potential CS majors without extensive *PriorExp* (who were likely first-years) as well as students in the Statistics and Neuroscience majors (who were likely non-first-years and have no *PriorExp*). As a result, these gaps likely do not reflect inequity in instruction.

Table 7. Summary of Existing Relationships between Demographic Variables and Metrics

Construct	Metric	Favored	Unfavored	CLES-Favored	Corrected p
<i>Performance</i>	Midterm 1	<i>Non-FirstGen</i>	<i>FirstGen</i>	0.69	0.0075
<i>Persistence</i>	Major Intent	<i>Year 1</i> <i>Some PriorExp</i>	<i>Year 2+</i> <i>No PriorExp</i>	0.80 0.66	5.3×10^{-10} 0.0077

Holm–Bonferroni corrections were performed across all tests for the same construct. For brevity, we only show effects that are significant at the 0.05 level post-correction. These effects are bolded. The CLES-Favored is the proportion of pairs of students in which the student in the “Favored” subgroup has a higher metric value. There were no significant KWH results. Thus, no post-hoc tests were performed.

Table 8. Summary of All-Population Results

Construct	Metric	CLES-I	Corrected p
<i>Acceptance</i>	See Table 4	0.556	0.31
<i>Affect</i>		0.599	0.07
<i>Desire to Participate</i>		0.448	0.34
<i>Membership</i>		0.562	0.25
<i>Trust</i>		0.591	0.09
<i>Self-Efficacy</i>		0.633	0.015*
<i>Performance</i>	Midterm 1	0.569	0.40
	Midterm 2	0.530	1
	Final	0.519	0.69
<i>Engagement</i>	Lab Attendance	0.458	0.71
	Lecture Attendance	0.480	0.66
<i>Persistence</i>	Major Intent Change	0.470	0.45

Holm–Bonferroni corrections were performed across all tests for the same construct. The only post-correction significant effect at the 0.05 level (*Self-Efficacy*) is bolded and followed by an asterisk (*). The CLES-I is the proportion of pairs of students in which the student in the intervention group has a higher metric value. Shaded effects are considered small but non-negligible by Vargha and Delaney’s guidelines [47] (see Table 6). We did not detect any medium or large effects for the entire population.

Moreover, we note that we only observe *Performance* gaps between *FirstGen* and *Non-FirstGen* students, i.e., we did not identify any significant relation between *Performance* and *Gender*, *Race*, *Latinx/e/a/o*, *Year*, and even *PriorExp*.³

6 Results

In this section, we begin by discussing population-wide treatment effects as summarized in Table 8, and then narrow down our analysis by demographic groups of interest.

³This does not mean that those performance gaps do not exist in our context. They may just not be strong enough to be detected by our analyses.

Table 9. Summary of Group Sizes, Intervention Effect Directions, and Intervention Effect Sizes by Demographic Subgroups

Construct	Metric N (control) N (intervention)	Gender		Race			Latinx/e/a/o		FirstGen		Year		PriorExp	
		H-Min	H-Maj	Asian	White	B/2+	Y	N	Y	N	1	2+	No	Some
		46	25	17	40	10	10	60	19	67	38	33	33	53
		37	20	22	21	8	15	41	19	54	29	27	23	50
<i>Acceptance</i>	See Table 4	.58	.60	.51	.62	.44	.56	.61	.65	.53	.60	.56	.62	.52
<i>Affect</i>		.57	.77	.45	.75	.52	.74	.62	.67	.58	.77	.46	.61	.59
<i>Desire to Participate</i>		.44	.43	.35	.43	.47	.43	.43	.40	.46	.41	.45	.55	.41
<i>Membership</i>		.52	.74	.38	.73	.55	.77	.55	.75	.52	.55	.62	.43	.62
<i>Trust</i>		.64	.55	.60	.65	.50	.57	.67	.76	.55	.64	.59	.57	.61
<i>Self-Efficacy</i>		.59	.73	.41	.70	.48	.82	.57	.66	.63	.70	.52	.65	.62
<i>Performance</i>	Midterm 1	.60	.52	.54	.60	.55	.64	.58	.57	.58	.60	.52	.57	.56
	Midterm 2	.56	.49	.43	.63	.46	.58	.56	.51	.54	.53	.52	.59	.50
	Final	.57	.47	.42	.63	.38	.54	.56	.53	.51	.60	.47	.61	.47
<i>Engagement</i>	Lab Attendance	.49	.34	.37	.48	.54	.35	.46	.46	.46	.41	.48	.48	.43
	Lecture Attendance	.55	.34	.40	.54	.57	.44	.48	.53	.47	.45	.51	.48	.47
<i>Persistence</i>	Major Intent Change	.44	.53	.40	.49	.65	.46	.47	.41	.49	.40	.57	.45	.48

Each cell shows the CLES-I for a metric (row) in a subpopulation (column). Effects of at least medium size (see Table 6) are shaded, and large effects are additionally bolded. Note that values smaller than 0.5 indicate a negative effect direction. The demographic data comes from two separate survey sources (see Section 5.1), so the *FirstGen* and *PriorExp* analyses involved more students. Not all students filled out every survey, so not every student is in every construct.

6.1 Intervention Effects on the Entire Student Population (RQ1)

Students from the intervention group had significantly larger increases in *Self-Efficacy* than students from the control group (CLES-I = 0.633, $p = 0.015$; see Table 8). There were no significant differences between the intervention and control groups on any other construct.

The effect sizes (CLES-I) were small (shaded) or negligible (non-shaded) for *all* constructs. However, the *direction* of the effects (i.e., whether each CLES-I is larger than .50) was mostly consistent with the findings from previous iterations of the EBI in the literature. Students from the intervention group had larger increases in all *Belonging* constructs (regarding the MWU tests) except *Desire to Participate*. They also had higher (but still non-significant) *Performance*, outscoring students from the control group in all three exams. In contrast, the direction of the effect sizes for the *Desire to Participate* and *Engagement* constructs favor students from the control group. Moreover, students in the control group had slightly higher *Persistence*, as measured by changes in their ITM in CS throughout the semester.

6.2 Intervention Effects on Demographic Subgroups (RQ2)

We next sought to understand whether our intervention benefited select demographic subgroups, like how Hammerlund et al.’s [21] intervention improved the performance of students who belong to historically underrepresented racial groups.

To examine whether our intervention had similar effects on individual demographic subgroups, we repeated our analysis in Section 6.1 after separating the student population by each demographic variable from Tables 1 and 2. Tables 9 and 10, respectively, summarize the effect sizes/directions and the significances of all tests.

For most constructs and demographic subgroups, the effects shown in Table 9 matched that of the whole population in Table 8. Additionally, among all *Belonging* effects that are medium or large, only one (*Desire to Participate* for Asian, CLES-I = .35, though insignificant; see Section 7) was in the negative direction. This implies that the intervention had mostly positive effects on all subgroups of students for those constructs. Furthermore, we observed many medium (CLES-I $\geq .64$) or even large ($\geq .71$) effects on *Belonging* constructs for H-Maj, White, Latinx/e/a/o, first-generation, and

Table 10. Summary of p-values for the Effects in Table 9

Construct	Metric	Gender		Race			Latinx/e/a/o		FirstGen		Year		PriorExp	
		H-Min	H-Maj	Asian	White	B/2+	Y	N	Y	N	1	2+	No	Some
<i>Acceptance</i>	See Table 4			0.94										
<i>Affect</i>					0.10		0.91	0.89			0.019*			
<i>Desire to Participate</i>						0.93								
<i>Membership</i>			0.36		0.18	0.86	0.61		0.39				0.55	
<i>Trust</i>		0.65			0.83			0.29	0.33				0.99	0.91
<i>Self-Efficacy</i>					0.38		0.31			0.39	0.26			0.55
<i>Performance</i>	Midterm 1													
	Midterm 2													
	Final													
<i>Engagement</i>	Lab Attendance													
	Lecture Attendance											0.95		
<i>Persistence</i>	Major Intent Change								0.91					

Holm–Bonferroni corrections were performed across all tests for the same construct. Post-correction $p = 1.0$ results are omitted for brevity. The only post-correction significant effect at the 0.05 level (*Affect* for first-year students) is shaded, bolded, and followed by an asterisk (*).

first-year students, although they were mostly insignificant due to the small sizes of these groups (see Section 7 for more discussion). For all of these findings, we note that the consistency in each row’s direction is not unexpected, as these demographic subgroups are not independent: each student appears at most⁴ six times in these comparisons, one for each demographic variable.

This analysis also revealed that first-year students in the intervention group had a significantly larger increase in *Affect* than first-year students in the control group (post-correction $p = 0.019$, large CLES-I = 0.77). This effect was not significant between experimental conditions for the whole class, which implies that the intervention was especially helpful in boosting first-year students’ *Affect* as they transition to college.

7 Discussion

7.1 Interactions between *Self-Efficacy* and Other Constructs

We found that students from the intervention group had significantly larger increases in *Self-Efficacy* than students from the control group (Section 6.1). This indicates that the EBI can improve students’ *Self-Efficacy*. Since *Self-Efficacy* has been linked to *long-term Performance* [35] and *Persistence* [9, 50], the EBI may also indirectly influence students’ *Performance* and *Persistence* over a longer period of time, though we did not see any immediate significant effects in our data.

Although the relationship between students’ *Self-Efficacy* and *Performance* is often considered bidirectional [7],⁵ our data suggests that *within the timeframe of the course*, it was more likely that *Performance* impacted *Self-Efficacy* than the other way around. This is because all three *Performance* metrics are significantly (rank-)correlated with students’ post-intervention *Self-Efficacy* (with weak relationships), but not pre-intervention *Self-Efficacy* (see Table 11). Moreover, the correlations were more pronounced for the two midterms than for the final. If *Self-Efficacy* had a stronger short-term impact on *Performance*, we would expect to see (1) the pre-intervention *Self-Efficacy* being more correlated to *Performance* than the post-intervention *Self-Efficacy*, and (2) the post-intervention *Self-Efficacy* (which was measured after both midterms but before the final) being more correlated to the final scores than the midterm scores. However, we note that EBI’s impact on *Self-Efficacy* *cannot* be fully explained by its impact on students’ *Performance*, as we only found the EBI to

⁴We say “at most” because students were not required to provide their gender, race, ethnicity, and class year in the surveys.

⁵Where higher *Performance* leads to higher *Self-Efficacy*, which then drives higher *Performance*.

Table 11. Spearman's Rank-Correlations between *Self-Efficacy* Metrics vs. *Performance* and *Persistence*

<i>Self-Efficacy</i> metric	<i>Performance</i>			<i>Persistence</i>		
	Midterm 1	Midterm 2	Final	Pre	Post	Change
Pre-intervention	.06	.13	.06	.16	.19	.08
Post-intervention	.39***	.36***	.29**	.10	.03	.11
Change	.34***	.24*	.22*	-.06	-.12	.08

The number in each cell is the Spearman's ρ between the column and row metrics. Results significant at the 0.05, 0.01, and 0.001 level (unadjusted) are followed by one (*), two (**), and three (***) asterisks, respectively.

Table 12. Spearman's Rank-Correlations between Students' *Self-Efficacy* Pre-Post Differences and Their *Belonging* Pre-Post Differences

<i>Belonging</i> construct (pre-post differences)	<i>Acceptance</i>	<i>Affect</i>	<i>Desire to Participate</i>	<i>Membership</i>	<i>Trust</i>
Spearman's ρ with <i>Self-Efficacy</i> pre-post differences	.51***	.53***	.34***	.45***	.50***

The number in each cell is the Spearman's ρ between the column and row metrics. All (unadjusted) p-values are lower than 0.001.

have a significant effect on the former (see Section 6.1). Furthermore, we note that *Self-Efficacy*'s short-term impact on students' *Persistence* is negligible in our data, see Table 11.

On the other hand, we found that students' pre-post *Self-Efficacy* differences were significantly (rank-)correlated with their pre-post differences in all five *Belonging* constructs, with the relationships being moderate ($\rho > .40$) for all constructs but *Desire to Participate* (see Table 12). This implies that the EBI's influence on *Self-Efficacy* may propagate to the *Belonging* constructs over time.

Lastly, we note that the direction of the effects for *Performance* and all but one *Belonging* construct favored students from the intervention group, and we saw a large and significant effect on the *Affect* construct for first-year students in favor of the intervention group (Section 6.2). It is possible that the EBI can significantly improve students' *Performance* and other kinds of *Belonging* than *Affect*, but that our sample sizes do not generate enough statistical power.

7.2 Context and Demographic Differences between Our Course and Prior Work

The literature on the EBI suggests that it is most effective in "chilly academic contexts" where negative and exclusionary stereotypes are present [10]. However, as we discussed in Section 3.2, our course does not necessarily fit that description. Also, much of the literature focuses on closing existing gender/race gaps, which we did not detect with significance in our classroom (Section 5.4). This may explain why there were no significant differences or large effects between the intervention and control groups for the vast majority of the constructs.

The above, in addition to the fact that we did not find any significant (or large) effects in favor of the control group, suggests that the EBI is mostly not harmful to students. However, we did observe a medium effect for *Desire to Participate* in favor of the control group for Asian students specifically (Section 6.2). We are unsure why this occurred. The other demographic variables were evenly distributed among Asian students in the intervention and control groups, i.e. there was no significant difference in the ratio of Asian students with and without *PriorExp* for the two groups,

and so on.⁶ Moreover, the number of Asian students who were in lab sections led by Asian TAs was similar for the two groups, so neither group would have benefited extra from shared identity with their TAs [30, 36]. Therefore, it is unlikely that other demographic variables and TA identity are related to this effect. More statistical power is likely needed to clarify its strength and direction.

Furthermore, as we discussed in Section 2.4, Binning et al. [10] found in classrooms with low gender diversity (<25% women), women who experienced the EBI had lower growth mindsets than women who had not. Although students' growth mindsets are not part of our study design (and therefore were not collected in the pre- and post-intervention instruments), the mid-semester survey that we used to collect demographic data (see Section 5.1) also included Dweck's 3-item growth mindset instrument [18]. We leveraged this data to examine whether a similar effect happened in our context. Analogous to our analyses in Sections 6.1 and 6.2, we used MWU tests to compare (1) the growth mindset levels of students in the intervention and control groups, and (2) the growth mindset levels of students between experiment conditions for each demographic group. None of these tests yielded significant results.⁷ Under the assumption that the groups had similar pre-intervention growth mindset levels, this additional analysis finds no evidence that the EBI had a negative effect on students' growth mindset. We note that this does not directly contradict Binning et al.'s findings, because they observed the effect in classrooms with low gender diversity, while our course has high gender diversity.

Lastly, we note that the high gender diversity of our course may be a reason why our study yielded relatively null findings, especially in contrast to prior studies that found that the EBI had positive and lasting outcomes. This work, therefore, also reaffirms that the effectiveness of the EBI is context-dependent [10]. Replications of our study will be helpful to better understand the nuances behind whether and when the EBI is beneficial for introductory CS students.

8 Limitations and Future Work

To the extent of our knowledge, this study is the first to adapt the EBI to an introductory CS course. Our findings suggest that the EBI may help improve students' self-efficacy and first-year students' *Affect*. However, the strength of the effects may be influenced by the course context. As we discussed in Section 3.2, our course does not reflect "chilly" academic contexts.⁸

Furthermore, our findings are limited in that our data is from a single offering of a medium-sized introductory CS course. First, the majority of the students in the course identify as White, *Non-FirstGen*, and not *Latinx/e/a/o*. We were unable to factor intersectionality into our analysis due to small sample sizes. Second, the students in the course mostly identify as being from the historical gender minority. This is not typical of many CS courses, and our findings may not be generalizable to CS courses with low gender diversity. Future work should study the EBI's effects in courses with chillier contexts, different student demographics, and larger enrollments.

Lastly, we did not have a standard control group and instead had students in the control group participate in an activity about students' study habits. We do not expect this activity to have any effect on students, particularly on constructs other than *Performance*. However, if the students had benefited from the control activity, then we would expect to have observed less differences

⁶We acknowledge that grouping all Asian students together when reporting demographics hinders and suppresses the racial, ethnic, and cultural diversity within all Asian students. We were unfortunately unable to mitigate this issue due to data collection and reporting restrictions.

⁷We note that this additional analysis was conducted independent of our main analyses in Sections 6.1 and 6.2 without applying any false-positive error rate control techniques. As such, these tests are sufficiently powered to identify potential negative relationships.

⁸As per Binning et al. [10], classrooms where students from a specific demographic group are assigned a lower status and questioned about their abilities and belonging.

between students in the experimental and control groups than between the experimental group and a hypothetical perfect control group in which students received no treatment. Assuming our control activity did not have *negative* impacts on students' *Self-Efficacy*, which has been established previously [12], the fact that we still found significant differences in students' change in *Self-Efficacy* and first-year students' change in *Affect* suggests that the EBI's effects on these constructs are real. Regardless, future replications of this experiment may consider using a different control activity.

9 Conclusion

Our research revealed that the EBI had a significant positive impact on students' self-efficacy and first-year students' sense of belonging (*Affect*). Although we did not see significant effects for other constructs, the directions of the effects mostly favored students who received the EBI. There were also no significant effects favoring students who did *not* receive the EBI. We therefore did not find evidence that the EBI was harmful to the students as a collective, although its impact on individual students may vary. However, more research is needed to explore the EBI's effects in other CS contexts.

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Appendix

A Detailed Intervention Structure

A.1 Activity Introduction

The TAs started off by introducing the purpose of the activity. Students in the intervention sections were told that the activity was meant to reassure first-years that the transition to college is hard and that everyone experiences feelings of self-doubt or feels overwhelmed. However, things get easier as they adjust to college. Students in the control sections were told that the activity was meant to help first-years get used to their college workload. Things may be hard at first, but will get easier as they develop good study habits.

A.2 Individual Writing and Reflection

Next, students were given 7–10 minutes to complete a writing task individually. During this time, the TAs sat at the front of the classroom and did not engage with students unless the students had questions about the task. Students in the intervention sections were asked to reflect on the experiences they have had since entering college. They were specifically told to write about the challenges they faced transitioning to college and how some of the difficulties and challenges have begun to improve over time. Students in the control sections were asked to reflect on what study habits have worked well (or not so well) for them in the past.

A.3 Reading Quotes from “Prior” Students

After students finished writing, the TAs passed out handouts that contained four quotes. These quotes were framed as being from students who had taken IntroPython previously and were read out loud by the TAs. In the intervention sections, the quotes were intended to instill the norm that adversity is normal, not unique to students from certain demographic groups, and surmountable. They discussed the common challenges that students face in the course, and how students overcame them. The quotes were written by the research team in Binning et al. [11], and modified where needed for our course context. An example quote is shown below:

“There were quite a few people in my [IntroPython] lab who had previous coding experience, so sometimes I felt a little embarrassed to ask questions. However, I quickly learned that other students usually had the same question I did, and we all benefited from working with each other and learning from each other. Sometimes I had difficulty with an idea that my classmates understood, and other times, they struggled with concepts that I understood. I remember there wasn't always an “aha!” moment, where everything clicked. It was usually much more gradual, with some concepts only becoming clear after lots of practice and discussion. I realized that

everyone struggles sometimes, and the important thing is to help each other out and not give up.”

The quotes for the control sections were modeled after those of the intervention sections. Although they also discussed specific challenges that students faced in the course, they did not emphasize the fact that these struggles were common or surmountable. Rather, they either discussed how the individual students overcame the challenges by improving their study habits or provided students with advice on how they can use course resources effectively. An example quote is shown below:

“I chose to take [IntroPython] [to fulfill a graduation requirement]. I’ve never been really good at math, and I didn’t take any CS courses in high school, so I was a little nervous. There are a lot of assignments in this course, and they all took me quite a while. I initially struggled to complete assignments and [short programming tasks] on time, and I had to use all of my late slips. However, I eventually learned to start assignments early and take advantage of office hours. The TAs were friendly, and I liked to go to office hours in person and get some help debugging my code. I would recommend showing up with some code already written though, because the help was more efficient that way, and I was able to spend less time in office hours.”

A.4 Small Group Discussion of Reflection and Quotes

After the TAs finished reading out the quotes, they asked students to form small groups of 3–4 students. Students were given questions to discuss in their groups, and encouraged to refer to what they had written down during their individual reflections. Students in the intervention sections were given the following questions presupposing the existence of struggle:

- What are some common themes across several of the stories we read?
- Why do you think that sometimes students do not realize that other people are also struggling with the course?
- How do you think the hardships you face will be different when you are a Junior or Senior?

Students in the control sections were told to discuss symmetric quotes about study habits:

- What are some common study habits that have helped other students?
- Why do you think that students sometimes struggle to develop good study habits?
- How do you think your study habits will be different when you are a Junior or Senior?

A.5 Whole Class Discussion

To end the activity, the TA brought the class back together and asked students to share what their group discussed with the rest of the class. In facilitating these discussions, the TA made references to their own experiences when appropriate. For the intervention sections, the TAs also emphasized the central message that struggles during the transition to college are normal and temporary.

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